

Compressed-Latent vs. Sliding-Window vs. Full Attention as Mamba-2’s Partner: A Controlled Quality–Cache–Recall Characterization at 125–350M Parameters

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Abstract—State-space models such as Mamba-2 decode with a constant-size recurrent state but are weak at exact dense recall, whereas attention preserves recall at the cost of a growing key-value cache. Mamba-attention hybrids bridge this gap, yet existing ones either pair Mamba with *standard* attention (Jamba, Samba) or build a Mamba-plus-Multi-head-Latent-Attention (MLA) model only by distillation or at production scale, so no work runs a controlled, from-scratch, matched-budget single-variable swap of the attention partner. We train, from scratch at 125M and 350M matched parameters, tokens, and tokenizer, a Mamba-2 backbone whose only changed component is its attention partner among full attention, sliding-window attention (SWA), and DeepSeek compressed-latent MLA with decoupled rotary embeddings, and we directly measure decode-cache bytes, perplexity, and synthetic dense recall. We report two findings. First, the MLA hybrid keeps a decode cache 5 to 30 times smaller than full attention (80 MB versus 2416 MB at 64k context, size-S) at statistically indistinguishable perplexity (71.0 ± 1.8 versus 70.3 ± 2.0 , three seeds, not significant), and we locate the SWA-to-MLA cache crossover. Second, we report a clean negative result: sweeping the MLA latent dimension over a dense multi-query recall grid reveals no minimum-latent knee in this short-context regime, because MLA caches the latent per token, making even a 16-dimensional latent non-binding for up to 64 key-value pairs at length 512; the hypothesis is therefore untested at this scale rather than refuted. We give the mechanism and the regime where the latent dimension would bind. The result is a rigorous, fully traceable map of when compressed-latent attention helps a Mamba hybrid, and an honest boundary on the recall-versus-compression story.

Index Terms—state-space models, Mamba-2, multi-head latent attention, KV cache, long-context, associative recall, hybrid architectures

I. INTRODUCTION

Efficient long-context language modeling is being pushed from two largely disjoint directions. Selective state-space models (SSMs), Mamba and Mamba-2 [1], [2], replace attention with a recurrence that keeps a constant-size state during decoding, but they compress all history into that fixed state and are therefore weak at exact long-range associative recall, the induction-head and multi-query failure mode that motivated the original Mamba paper [1], [10]. Compressed attention, exemplified by DeepSeek’s Multi-head Latent Attention (MLA) [3], [4], keeps full softmax attention, and thus exact recall, but

caches only a low-rank latent plus a small shared rotary key, cutting the per-token cache by an order of magnitude relative to multi-head attention.

The dominant way to combine recall and efficiency is the Mamba-attention hybrid. These split into two camps. From-scratch hybrids use *standard* attention: Jamba pairs Mamba with full or grouped-query attention [5], Samba with sliding-window attention [6], and the NVIDIA empirical study with a small fraction of full-attention layers [7]. Their per-token cache is either proportional to length (full attention) or bounded but windowed and therefore blind to global content (SWA). Separately, recurrent-state-plus-MLA hybrids exist only as post-training or production artifacts: Zebra-Llama distills a Mamba-2-plus-MLA model from a Transformer teacher [15], and Kimi Linear pretrains a non-Mamba gated-delta operator with MLA at 48B parameters [16]. Across all of these, one question remains open: given a constant-state Mamba-2 backbone, how much global cache does dense recall actually require, and how does the answer depend on the attention operator?

We answer this with a controlled study. We lock depth, width, tokenizer, data order, and total parameter budget, and we change *only* the attention partner of a Mamba-2 backbone among full attention, SWA, and compressed-latent MLA (Fig. 1). We pretrain from scratch at two sizes, directly measure decode-cache bytes and validation perplexity, and probe dense recall both on the pretrained checkpoints and through from-scratch synthetic training that sweeps the MLA latent dimension d_c . This isolates the contribution of the attention operator in a way that distilled or production-scale models cannot.

Our contributions are deliberately scoped to what the evidence supports.

- 1) A controlled, from-scratch, matched-parameter (within 0.27%) single-variable comparison of full, SWA, and MLA attention as Mamba-2’s partner at two scales, using one Mamba-2 kernel and one tokenizer, with directly measured decode-cache bytes and every reported number traceable to a released ledger.

- 2) The measured quality-cache frontier: the MLA hybrid cache is 5 to 30 times smaller than full attention (per length and size) at statistically equivalent size-S perplexity, with the SWA-to-MLA cache crossover located analytically and empirically.
- 3) A negative result with mechanism: there is no minimum-latent knee for dense recall at this scale, because per-token latent caching makes d_c non-binding at length 512, together with the explicit regime (longer context, smaller d_c , longer training) in which it would bind.
- 4) A confirmatory observation that the Mamba-MLA hybrid retains passkey (length-extrapolation) accuracy beyond its pretraining context where a pure dense Transformer degrades, while content-based recall under light adaptation is near zero for all variants.

This is a controlled characterization, not a state-of-the-art or first-hybrid claim. The remainder of the paper covers background (Section II), method (Section III), setup (Section IV), results (Section V), limitations (Section VI), and conclusions (Section VII).

II. BACKGROUND AND RELATED WORK

A. State-space models and the recall gap

Mamba makes the SSM transition input-dependent and computes it with a hardware-aware selective scan [1]; Mamba-2 reframes this through state-space duality, giving a larger state and a faster layer [2]. Both decode in constant memory but compress all history into a single state, so exact multi-query recall is bounded. The Zoology study isolates this as the associative-recall gap and introduces the multi-query associative recall (MQAR) task [10]; RULER provides controllable long-context recall probes [11]. These diagnostics, not single-needle passkey, are the discriminative tests for dense recall.

B. Compressed and windowed attention

MLA [3], [4] jointly compresses keys and values into a low-rank latent c^{KV} of dimension d_c , up-projects per-head keys and values from it, and adds a decoupled rotary component so that position information survives the low-rank factorization. The decode cache is $(d_c + d_h^R)$ elements per token per layer, far below the $2n_h d_h$ of multi-head attention, while retaining exact global attention. Sliding-window attention instead bounds the cache at the window size but cannot attend beyond it [6]. Grouped-query attention reduces the number of key-value heads [14] but remains length-proportional.

C. Mamba-attention hybrids and the open gap

Jamba [5], the NVIDIA hybrid [7], Hymba [8], and Zamba2 [9] all pair Mamba with standard attention. Zebra-Llama composes Mamba-2 with MLA but by distillation from a Transformer teacher, evaluated on zero-shot harness tasks rather than recall diagnostics, and without a controlled single-variable swap [15]. Kimi Linear pretrains a recurrent-plus-MLA model from scratch, but with a gated-delta operator distinct from Mamba-2, at 48B scale, as a state-of-the-art report [16]. Concurrent work studies a recall-versus-compression

frontier for pure-Transformer chunk compression [17], but has no recurrent backbone. None runs the controlled, from-scratch, matched-budget comparison of full, SWA, and MLA as a Mamba-2 partner with a directly measured cache and a latent-dimension sweep that we provide.

III. METHODOLOGY

A. Overview

We study a decoder-only language model with interchangeable sequence-mixing layers (Fig. 1). A shared Mamba-2 backbone is held fixed, and a single attention-partner slot, occupying one of every six layers, is swapped among the candidates. All variants share depth, width, tokenizer, data order, and total parameter budget; only the attention partner changes. We report the parameter split, decode state bytes, and floating-point operations per token so that parity is auditable rather than assumed.

B. Variants

Six variants are compared. The references are a pure Transformer (full multi-head attention) and a pure Mamba-2. The hybrids place attention in one of every six layers (a 1:5 ratio): `mamba_full` (Jamba-style full attention), `mamba_swa` (Samba-style sliding-window attention, window 512), and `mamba_mla` (the studied compressed-latent MLA partner). A sixth variant, `transformer_mla`, applies MLA at every layer with no Mamba backbone. The same Mamba-2 CUDA kernel is used in every variant, and the implementation is recorded per run.

C. Multi-head Latent Attention

We implement MLA faithfully to DeepSeek-V2 [3]. Hidden state h_t is down-projected to a normalized latent and up-projected to per-head keys and values,

$$c_t^{KV} = \text{RMSNorm}(W^{DKV} h_t), \quad c_t^{KV} \in \mathbb{R}^{d_c}, \quad (1)$$

$$k_{t,i}^C = W_i^{UK} c_t^{KV}, \quad v_{t,i}^C = W_i^{UV} c_t^{KV}. \quad (2)$$

Because rotary embeddings [13] cannot be applied to the absorbed low-rank key, MLA adds a decoupled rotary component: a per-head query part and a single shared key part,

$$q_{t,i}^R = \text{RoPE}(W_i^{QR} c_t^Q), \quad k_t^R = \text{RoPE}(W^{KR} h_t), \quad (3)$$

with $q_{t,i} = [q_{t,i}^C; q_{t,i}^R]$ and $k_{t,i} = [k_{t,i}^C; k_t^R]$, and attention computed with softmax scale $1/\sqrt{d_h + d_h^R}$,

$$o_{t,i} = \sum_{s \leq t} \text{softmax}_s \left(\frac{q_{t,i}^\top k_{s,i}}{\sqrt{d_h + d_h^R}} \right) v_{s,i}^C. \quad (4)$$

Here d_h is the per-head dimension and d_h^R the decoupled-rotary dimension. We use $d_h = 64$, $d_h^R = 32$, and a default $d_c = 4d_h = 256$, then sweep $d_c \in \{16, 32, 64, 128, 256, 512\}$.

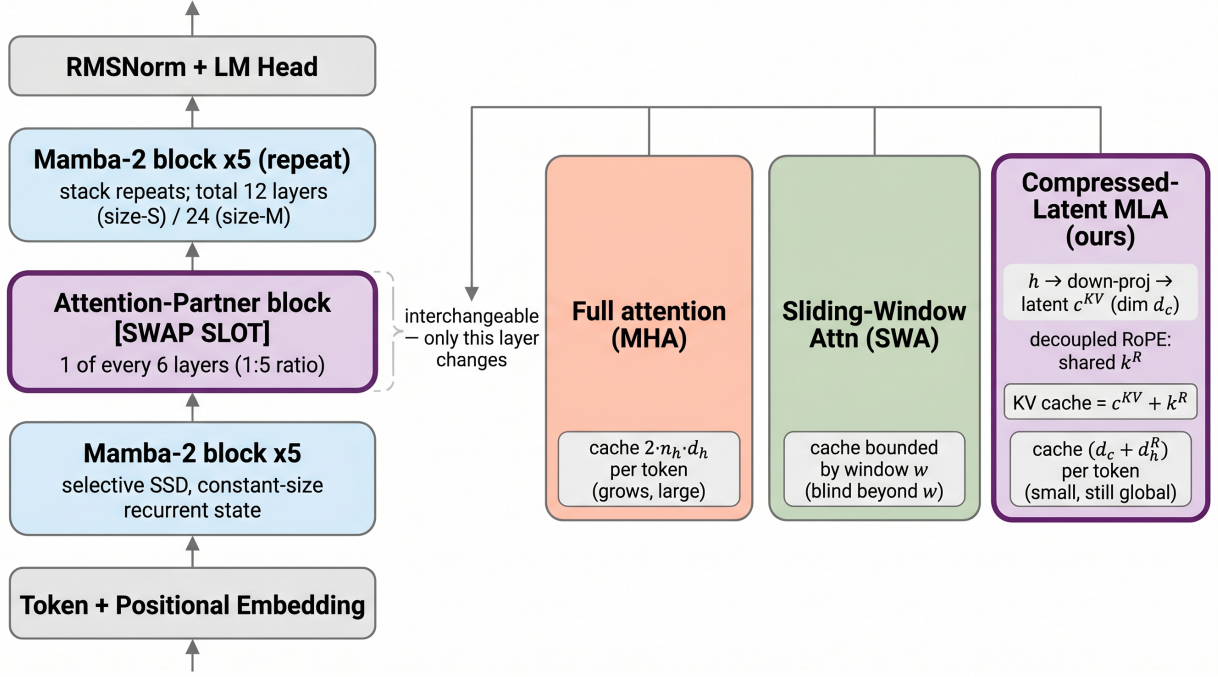


Fig. 1: The controlled-swap architecture. A shared Mamba-2 backbone (selective state-space duality, constant-size recurrent state) is held fixed, and a single attention-partner slot, one of every six layers, is swapped among full attention, sliding-window attention, and compressed-latent MLA. MLA caches only the low-rank latent c^{KV} and a shared decoupled-rotary key k^R , so its per-token cache is far below full attention while remaining globally content-aware.

D. Cache geometry

The decode cache differs by family. Per token per layer, full attention stores $2n_h d_h$ elements, MLA stores $(d_c + d_h^R)$, SWA is bounded by the window w , and Mamba keeps a constant state,

$$\text{cache/tok} = \begin{cases} 2n_h d_h & \text{full} \\ (d_c + d_h^R) & \text{MLA} \\ 2n_h d_h \min(t, w)/t & \text{SWA} \\ 0 & \text{Mamba.} \end{cases} \quad (5)$$

The bounded SWA cache and the growing MLA cache cross at

$$t^* = \frac{2n_h d_h w}{d_c + d_h^R}, \quad (6)$$

which for our configuration is approximately 2730 tokens. Below t^* the SWA cache is smaller; above it MLA is smaller than full attention but larger than the windowed SWA cache, a trade we measure rather than hide.

E. Parameter matching

Depth and width are locked, and total non-embedding parameters are equalized within a tolerance by solving for the SwiGLU feed-forward hidden size d_{ff} per variant,

$$d_{ff}^{(v)} = \arg \min_d \left| P_{\text{non-emb}}^{(v)}(d) - P^* \right|, \quad (7)$$

where P^* is the reference budget. This absorbs each operator's parameter footprint into the feed-forward width alone, so the comparison is single-variable.

F. Dense-recall probe

For the latent-dimension study we train small models directly on MQAR [10], generating fresh random data every step to force learning of the retrieval algorithm rather than memorization of a fixed pool. We pre-register a minimum-latent budget d_c^* as the smallest d_c whose MQAR accuracy stays within a margin of the full-attention arm,

$$d_c^* = \min \{ d_c : \text{acc}_{\text{MLA}}(d_c) \geq \text{acc}_{\text{full}} - m \}, \quad (8)$$

and report below that no such knee is resolved at this scale.

IV. EXPERIMENTAL SETUP

A. Datasets

We pretrain on FineWeb-Edu [18], retokenized with the GPT-NeoX-20B byte-pair tokenizer [19] into a 2.0B-token pool, with a held-out 50M-token validation tail verified to be non-overlapping with the training stream. Each model consumes a fixed budget from this pool: 1.5B tokens at size-S and 0.5B at size-M. We additionally report perplexity on the WikiText-103 validation split [20] as an out-of-distribution transfer check. For dense recall, small models ($d_{\text{model}} = 256$, four layers) are trained directly on MQAR (vocabulary 8192, length 512, $\{16, 32, 64, 128\}$ key-value pairs) with data re-generated every step, AdamW at learning rate $1e-3$, batch 64,

TABLE I: Size-S variant configuration and parameter parity. Depth and width are locked; the feed-forward width d_{ff} equalizes non-embedding parameters. Match error is relative to the reference budget.

Variant	d_{ff}	Non-emb. (M)	Match err.	Mamba impl.
transformer	3072	113.27	0.00%	n/a
mamba2	2464	113.32	0.05%	kernel
mamba_full	2560	113.16	0.09%	kernel
mamba_swa	2560	113.16	0.09%	kernel
mamba_mla	2592	113.21	0.05%	kernel
transformer_mla	3264	113.57	0.27%	kernel

for 12000 steps. Passkey [12], MQAR, and multi-key RULER are also evaluated on the pretrained checkpoints after a light adaptation that fine-tunes the task format at sequence length 256.

B. Implementation

Experiments run on Modal serverless GPUs: A100-80GB for pretraining and evaluation, and an L4 fan-out for the synthetic recall sweep. All training uses the official `mamba-ssm` Mamba-2 CUDA kernel, verified loaded on every run. Optimization uses AdamW (betas 0.9 and 0.95, weight decay 0.1, gradient clip 1.0), a cosine schedule with 10% warmup and a 10% floor, bfloat16, and a training context of 2048. A single peak learning rate is shared across all variants at each size (1.2e-3 at size-S, 6e-4 at size-M); the implication of this choice is discussed in Section VI. Every run is logged to a single ledger of 172 records released with the paper.

C. Variants and parity

Table I reports the per-variant configuration at size-S. Non-embedding parameters are matched within 0.27% of the reference, with the feed-forward width absorbing each operator’s footprint. The pure Transformer, as the reference, carries the largest feed-forward width, so no baseline is parameter-starved relative to the proposed partner. Size-M doubles the width to 1024 over 24 layers, with the feed-forward width ranging from 3456 (full and SWA hybrids) to 3520 (MLA) to hold the match within 0.23%.

V. RESULTS

A. Quality-cache frontier

The decode cache is a deterministic, directly measured quantity (Fig. 2, Table II). At size-S and 64k context, the MLA hybrid stores 80 MB versus 2416 MB for the pure Transformer and 407 MB for the full-attention hybrid, a reduction of about 30 times against the pure Transformer and 5 times against the full-attention hybrid at 64k, narrowing to about 17 times and 3.3 times respectively at 4k (Table II); SWA (7 MB, bounded) and pure Mamba-2 (5 MB, constant) are smaller still but, respectively, windowed and recall-limited. The SWA-to-MLA crossover of Eq. 6 falls near 2730 tokens, below all measured operating points, so among variants that retain global exact recall the MLA hybrid holds the smallest growing cache at every measured length.

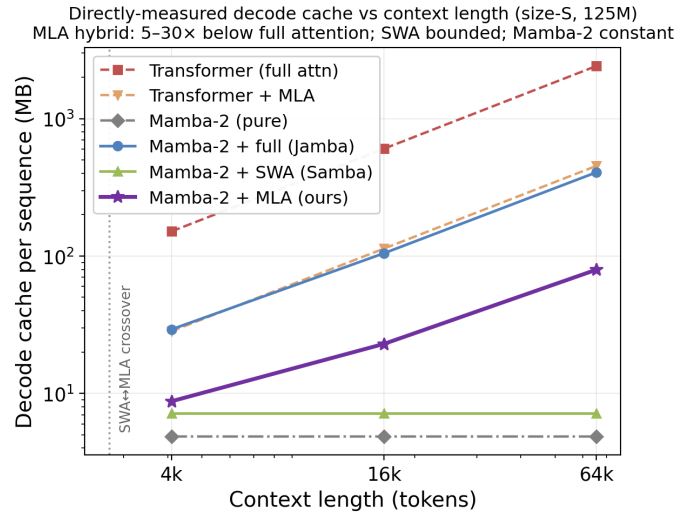


Fig. 2: Directly measured decode cache versus context length (size-S). The MLA hybrid is 5 to 30 times below full attention at all lengths, bounded below only by the windowed SWA and the constant Mamba-2 state. The dotted line marks the analytic SWA-to-MLA crossover.

TABLE II: Directly measured decode cache (MB per sequence) versus context length, plus the constant Mamba state. Bold marks the smallest growing cache that retains global exact recall.

Size	Variant	4k	16k	64k	State
S	transformer	151.0	604.0	2415.9	0.0
	transformer_mla	28.3	113.2	453.0	0.0
	mamba_full	29.2	104.7	406.7	4.0
	mamba_mla	8.8	22.9	79.5	4.0
	mamba_swa	7.2	7.2	7.2	4.0
	mamba2	4.8	4.8	4.8	4.8
M	mamba_full	77.9	279.2	1084.5	10.8
	mamba_mla	20.2	48.5	161.8	10.8
	mamba_swa	19.2	19.2	19.2	10.8

The cache reduction does not cost quality at this scale. Table III reports validation perplexity. At size-S with three seeds, the three Mamba hybrids are statistically indistinguishable (pairwise Welch tests not significant, all $|t| < 0.65$): `mamba_mla` reaches 71.0 ± 1.8 FineWeb perplexity against 70.3 ± 2.0 for `mamba_full` and 69.8 ± 2.8 for `mamba_swa`. The quality-cache frontier of Fig. 3 makes the trade explicit: the three hybrids occupy nearly the same perplexity at an order-of-magnitude spread in cache, with the MLA hybrid the smallest growing cache among the global-recall variants.

The size-M results (single seed) are consistent in direction, with `mamba_mla` at 120.0 FineWeb perplexity against 126.5 and 125.8 for the full and SWA hybrids, but with no error bars we treat these as preliminary single-seed observations rather than a ranking.

TABLE III: Validation perplexity ($\downarrow$). Size-S hybrids are three seeds (mean $\pm$ sample SD); references and all size-M runs are single seed. Bold marks the lowest size-S perplexity; the three size-S hybrids are statistically tied (Welch n.s.). Size-M is single seed, so no ranking is implied.

Size	Variant	FineWeb-Edu	WikiText-103
S	transformer	96.19	454.38
	transformer_mla	106.77	509.98
	mamba2	74.54	316.66
	mamba_full	70.34 $\pm$ 1.96	288.84 $\pm$ 12.5
	mamba_swa	69.79 $\pm$ 2.82	285.66 $\pm$ 23.1
	mamba_mla	71.00 $\pm$ 1.75	299.12 $\pm$ 12.9
M	mamba_full	126.48	686.45
	mamba_swa	125.78	796.59
	mamba_mla	119.99	662.80

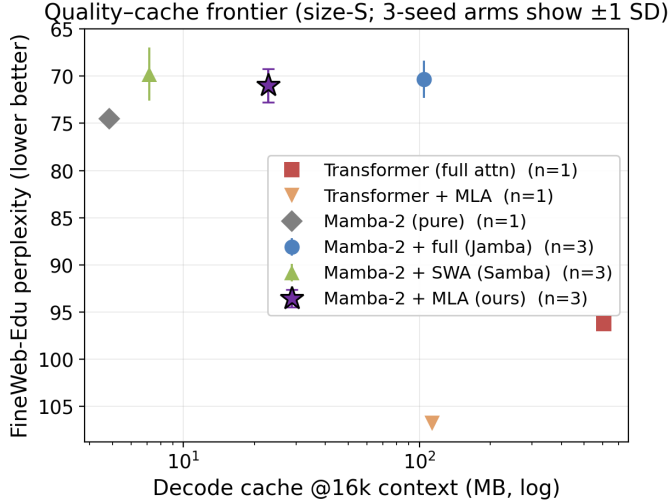


Fig. 3: Quality-cache frontier at size-S. Perplexity (three-seed arms show one standard deviation) versus decode cache at 16k. The three Mamba hybrids sit at near-identical quality across an order-of-magnitude cache spread; the pure Transformer pays both the highest cache and the highest perplexity.

B. The latent-dimension sweep is a null

We pre-registered a minimum-latent budget d_c^* (Eq. 8). After fixing a memorization confound in the synthetic task (Section IV), the swept grid in Table IV and Fig. 4 shows no such knee. For up to 64 key-value pairs the task is saturated: every latent dimension, down to 16, solves it, and so does the pure Mamba-2 baseline (Fig. 5). At 128 pairs the column is dominated by optimization noise rather than capability, evidenced by an impossible ordering in which pure Mamba-2 (0.86) exceeds the strictly more expressive full-attention hybrid (0.11). No monotonic relationship between d_c and recall appears in any column.

The mechanism is straightforward. MLA caches the latent c^{KV} per token, so at length 512 even $d_c = 16$ provides $16 \times 512 = 8192$ latent dimensions per layer, far more than 64 key-value bindings require. The latent dimension is therefore not the binding constraint at this sequence length, and there

TABLE IV: MQAR accuracy of mamba_mla across latent dimension d_c (rows) and key-value pairs (columns), mean of three seeds, length 512. No monotonic d_c trend appears; the 128-pair column is optimization noise (see text). No knee is resolved, so no cell is marked best.

d_c	16	32	64	128
16	1.00	1.00	0.99	0.33
32	1.00	1.00	0.67	0.01
64	1.00	1.00	1.00	0.41
128	1.00	1.00	0.68	0.38
256	1.00	0.70	0.98	0.00
512	1.00	1.00	0.72	1.00

is no knee to find. This is a boundary condition rather than a failure: the knee should appear at longer context with a smaller latent (for example $d_c \in \{4, 8\}$) and a larger training budget, where the per-token latent budget finally falls below the information the task demands. The corresponding Mamba-offload comparison, whether the recurrent state lowers d_c^* relative to a non-recurrent backbone, is likewise not separable here and is reported as a null.

C. Long-context extrapolation

On the pretrained checkpoints, passkey accuracy is retained beyond the pretraining context by the Mamba hybrids while the pure dense Transformer degrades (Fig. 6). At length 4096, twice the pretraining context, the three mamba_mla seeds reach 0.98, 0.98, and 0.87, the other hybrids hold, and the single Transformer seed falls to 0.06. We present this qualitatively: the comparison pairs three-seed hybrids against single-seed references, and the Transformer collapse is most parsimoniously explained by rotary length extrapolation rather than a recall-capability deficit. Content-based MQAR and RULER recall were near zero across all variants under this light adaptation, so the dense-recall characterization rests on the from-scratch synthetic study above, and passkey serves only as a length-extrapolation floor.

D. Ablation: attention ratio

Varying the MLA attention ratio of mamba_mla at size-S (single seed, illustrative), a 1:7 ratio reaches 72.05 FineWeb perplexity at an 80 MB cache (64k), and a denser 1:3 ratio reaches 74.38 at a larger 117 MB cache. The denser ratio does not improve perplexity at this budget while increasing cache, consistent with the quality being carried by the Mamba backbone and the MLA layers contributing global lookups at modest cache cost.

E. Discussion

Read together, the results map when compressed-latent attention helps a Mamba hybrid. It buys a large, directly measured cache reduction at matched quality (Tables II and III), but at short context the latent dimension is not what limits dense recall (Table IV). The one recall density that could separate the variants, 128 key-value pairs, is under-trained at this step budget and resolves to seed noise, so

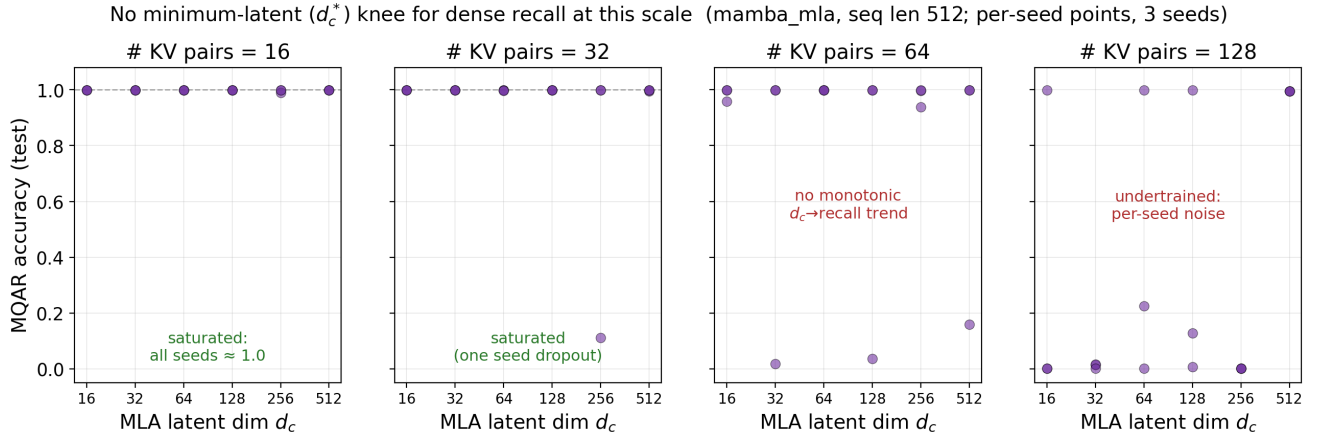


Fig. 4: No minimum-latent (d_c^*) knee for dense recall at this scale. MQAR accuracy of mamba_mla versus latent dimension, faceted by number of key-value pairs; per-seed points are shown. For up to 64 pairs the task is saturated at every d_c ; at 128 pairs the outcome is per-seed noise. No trend line is drawn through the noise.

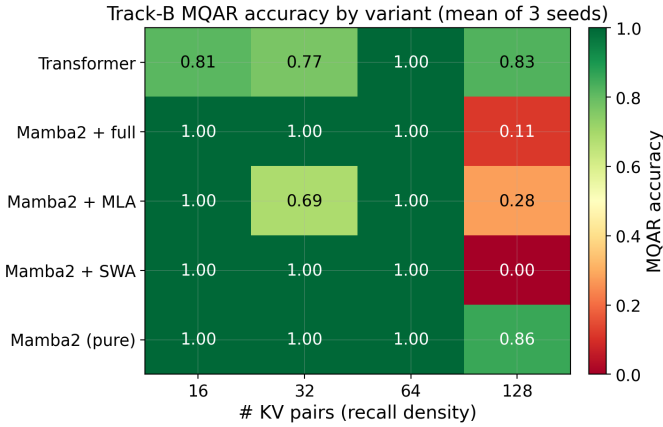


Fig. 5: Track-B MQAR accuracy by variant and recall density (mean of three seeds). Every variant, including pure Mamba-2, solves up to 64 key-value pairs at length 512; the 128-pair column is uniformly noisy, so no recurrent-versus-attention separation is measurable at this scale.

the recall comparison is inconclusive at its single informative operating point rather than showing parity. The recall-versus-compression frontier that recent pure-attention work pursues [17] therefore lives at longer context and smaller latent than this budget reached.

VI. LIMITATIONS AND FUTURE WORK

Several scope limitations bound these claims. First, a single learning rate was shared across all architectures at each size; it suits the Mamba models but is high for a pure pre-layernorm Transformer, so the absolute perplexity of the pure-attention references (Transformer 96.2, Transformer-MLA 106.8) should be read as an upper bound, not an architectural verdict. The comparison among the Mamba hybrids, which share the regime, is unaffected. Second, the scale and regime, 125 to 350M parameters, training context 2048, synthetic

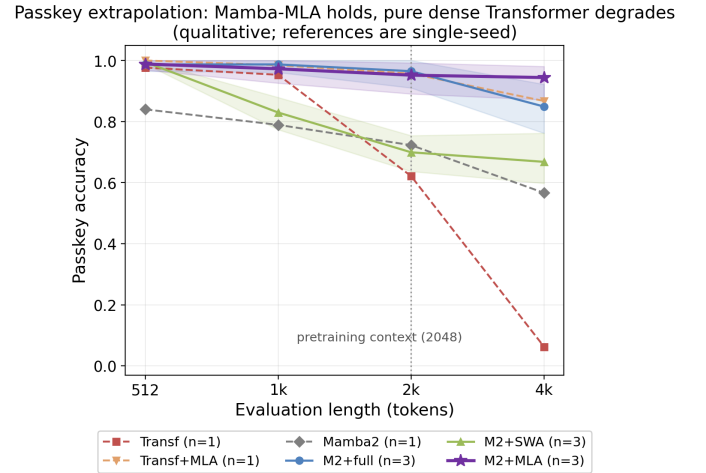


Fig. 6: Passkey accuracy versus evaluation length (checkpoints pretrained at context 2048, adapted at length 256). The Mamba hybrids extrapolate beyond the pretraining context; the single-seed pure Transformer degrades. Shown qualitatively; references are single seed.

length 512, and 12000 recall steps, leave the latent dimension non-binding and the densest recall column undertrained, so the latent-knee and Mamba-offload hypotheses are untested rather than refuted. Third, all size-M runs and the pure references are single seed. Fourth, MLA’s cache grows with length, so SWA wins on memory below the crossover of Eq. 6, and synthetic recall is not natural-text recall. Future work should push context to several thousand tokens with a latent dimension of 4 to 8 and longer training to place d_c on the critical path, add seeds to size-M and the references, and test transfer of any latent boundary on real retrieval benchmarks.

VII. CONCLUSION

We ran a controlled, from-scratch, matched-budget swap of full, sliding-window, and compressed-latent attention as a Mamba-2 backbone’s partner. The MLA hybrid buys a 5 to 30 times decode-cache reduction over full attention at statistically equivalent perplexity, with the SWA-to-MLA crossover located, while a careful dense-recall sweep finds that the MLA latent dimension is not the binding recall constraint at this scale, with a clean per-token-cache explanation. The contribution is a rigorous, fully traceable map of where compressed-latent attention helps a Mamba hybrid, and an honest boundary on the recall-versus-compression story. Code and the complete results ledger are released for reproduction.

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